

Supplementary material

Healthy-Data-Only, Gradient-Free On-Device Recalibration of an Integer-Only Vibration

Fault Classifier on a 20 KB-RAM Microcontroller

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This document contains material that supports the paper but was moved out of the main text to meet the length limit of the journal. All numbers are produced by the released scripts from the result files of the five-seed runs; $\pm$ denotes the population standard deviation.

S1. Validation of the theory on synthetic data

To test Propositions 1–3 and the integer implementation under a known shift, three-class 16 kHz vibration signals were synthesised: the healthy class contains shaft harmonics and noise, the outer-race class adds periodic decaying impulses, and the inner-race class adds impulses whose amplitude is modulated at the shaft frequency. Five surrogate models with the architecture of Table 3 of the paper were built with random convolution weights and a ridge-regression output layer, and quantised to int8. The shift was applied in the first layer by multiplying its weights by γ and replacing its bias by $\gamma b + \delta$. The shifted weights are held as wider integers (beyond the int8 range for $\gamma > 1$), so this is an idealised check of the derivation and the code, not of int8 deployment. With integer γ and δ the shift is exactly affine; with continuous $\gamma \in [0.5, 2]$, weight rounding makes it only approximately affine. These results validate the theory and the code and say nothing about diagnostic performance on real bearings.

Table S1. Validation on synthetic data (5 surrogate models, mean $\pm$ std, %).

Setting	Method	Accuracy	Agreement with source model
No shift, no calibration (source model)	–	82.4 $\pm$ 3.5	100.0
Exact affine shift ($\gamma \in \{1, 2, 3, 4\}$)	M0	33.7 $\pm$ 0.7	34.7 $\pm$ 1.4
Exact affine shift, exact statistics	M3L1	82.4 $\pm$ 3.4	99.7 $\pm$ 0.3
Approximate affine shift ($\gamma \in [0.5, 2]$)	M0	41.6 $\pm$ 15.5	41.6 $\pm$ 15.5
Approximate affine shift, $N = 32$	M1	47.2 $\pm$ 3.0	52.6 $\pm$ 6.0
Approximate affine shift, $N = 32$	M3	79.2 $\pm$ 3.6	86.4 $\pm$ 3.7
No shift, exact statistics	M1	49.6 $\pm$ 2.2	54.4 $\pm$ 5.4
No shift, exact statistics	M3	82.4 $\pm$ 3.5	100.0 $\pm$ 0.0
No shift, $N = 32$	M1	46.1 $\pm$ 2.6	51.7 $\pm$ 5.8
No shift, $N = 32$	M3	79.5 $\pm$ 4.0	86.9 $\pm$ 3.4
$\gamma = 6$ (beyond the clip limit of 4)	M0 / M3L1	33.3 $\pm$ 0.0 / 52.4 $\pm$ 1.6	33.9 $\pm$ 1.3 / 53.9 $\pm$ 2.1

Under an exact affine shift, recalibration restores the agreement with the source model from 34.7% to 99.7%, the remainder being due to fixed-point rounding (Proposition 1). Without a shift and with exact statistics, the healthy anchor leaves the model unchanged, whereas the all-class anchor lowers accuracy from 82.4% to 49.6% (Proposition 2). With 32 windows per layer, estimation noise lowers the agreement of M3 to about 87%. When fault windows are mixed into the calibration data, the error of the estimated mean of an intermediate layer grows linearly with ϵ ($R^2 = 0.998$, Figure S1a), as predicted by Proposition 3; this checks the implementation rather than providing independent evidence. Accuracy hardly changes for $\epsilon \leq 0.1$ and falls to 57.5% at $\epsilon = 0.5$ (Figure S1b). A shift beyond the clipping range is only partly recovered.

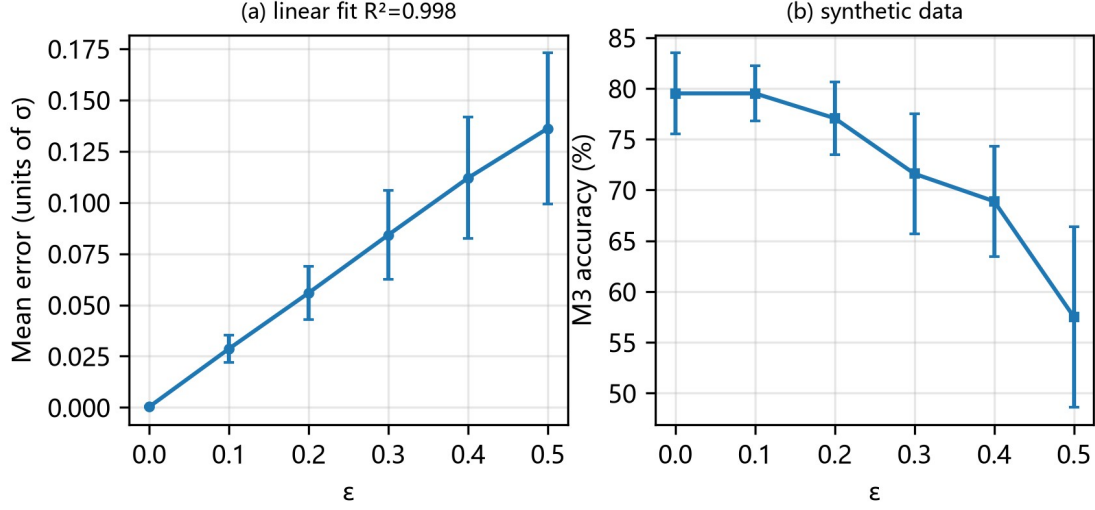


Figure S1. Proposition 3 on synthetic data: (a) fraction ϵ of fault windows versus the error of an intermediate-layer mean (in units of the healthy standard deviation); (b) ϵ versus M3 accuracy. Mean $\pm$ std over five surrogate models.

S2. Sensitivity of α -BN to α

Table S2. Accuracy of α -BN for different α (% , mean over 5 seeds); α is selected by the highest development mean.

Target	$\alpha = 0.05$	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$	$\alpha = 0.5$	M3L1
C1 (development)	27.7	31.2	31.4	29.7	27.5	11.9
C2-1HP (development)	84.8	74.0	54.1	44.8	41.3	88.4
P1 (development)	68.9	84.8	86.7	78.0	70.2	82.3
Development mean	60.5	63.4	57.4	50.8	46.3	60.9
C2-2HP (held out)	80.0	71.2	52.2	42.8	39.4	84.5
C2-3HP (held out)	76.4	68.7	53.1	43.8	40.4	83.3
P2 (held out)	99.2	96.4	85.2	77.3	73.5	99.8
P3 (held out)	83.2	84.0	74.1	64.9	60.0	80.0
Held-out mean	84.7	80.1	66.1	57.2	53.3	86.9

S3. Macro-F1

Table S3. Macro-F1 on each target (% , mean $\pm$ std over 5 seeds).

Target	M0	M1	M3	M3L1	B2*	M4
C1	13.4 $\pm$ 3.2	21.0 $\pm$ 3.5	4.9 $\pm$ 3.4	6.0 $\pm$ 2.6	24.7 $\pm$ 4.5	93.0 $\pm$ 1.5
C2-1HP	89.1 $\pm$ 3.6	30.6 $\pm$ 12.3	71.2 $\pm$ 12.1	86.6 $\pm$ 5.1	69.6 $\pm$ 7.9	98.6 $\pm$ 0.4
C2-2HP	86.1 $\pm$ 5.3	28.5 $\pm$ 12.0	69.2 $\pm$ 17.0	81.1 $\pm$ 7.7	65.4 $\pm$ 7.0	98.7 $\pm$ 0.5
C2-3HP	80.4 $\pm$ 5.4	29.4 $\pm$ 12.6	55.4 $\pm$ 16.5	80.7 $\pm$ 7.9	62.1 $\pm$ 4.9	98.8 $\pm$ 0.4
P1	39.6 $\pm$ 11.0	66.7 $\pm$ 3.3	77.8 $\pm$ 6.1	82.0 $\pm$ 4.7	84.3 $\pm$ 4.2	96.7 $\pm$ 0.3
P2	99.8 $\pm$ 0.1	69.6 $\pm$ 2.6	99.8 $\pm$ 0.1	99.8 $\pm$ 0.1	96.4 $\pm$ 1.0	99.9 $\pm$ 0.1
P3	73.2 $\pm$ 5.6	57.6 $\pm$ 3.2	75.8 $\pm$ 6.6	78.9 $\pm$ 6.4	83.7 $\pm$ 2.7	99.4 $\pm$ 0.2
Mean	68.8	43.3	64.9	73.6	69.5	97.9

S4. Additional figures

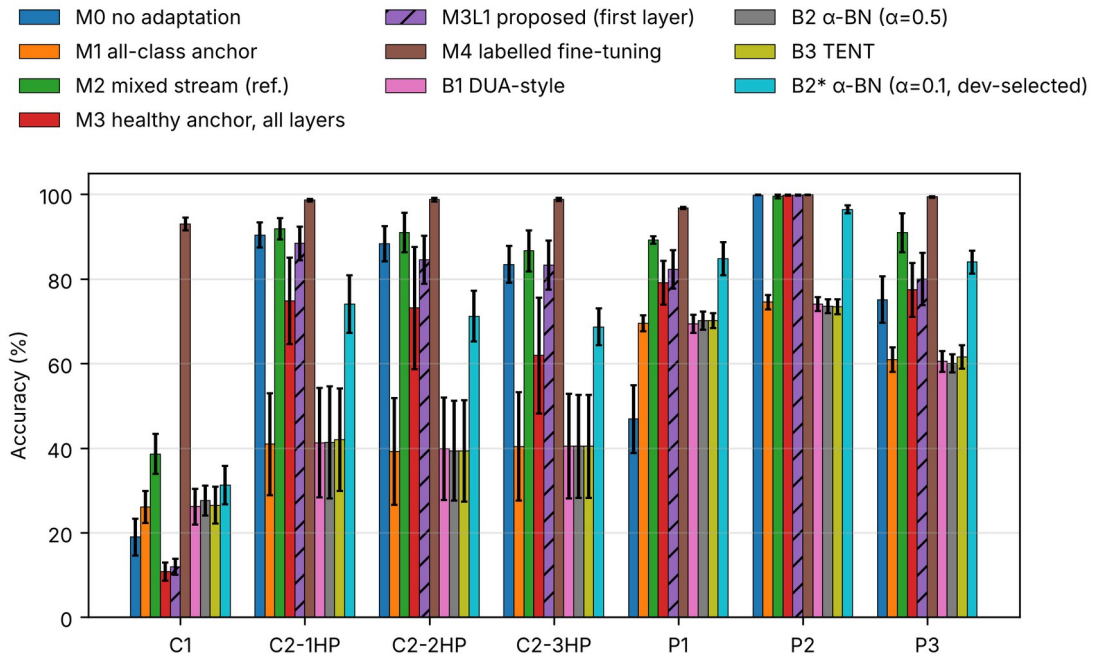


Figure S2. Accuracy of all methods on the seven target domains (mean $\pm$ std over 5 seeds); the data are those of Table 5 of the paper.

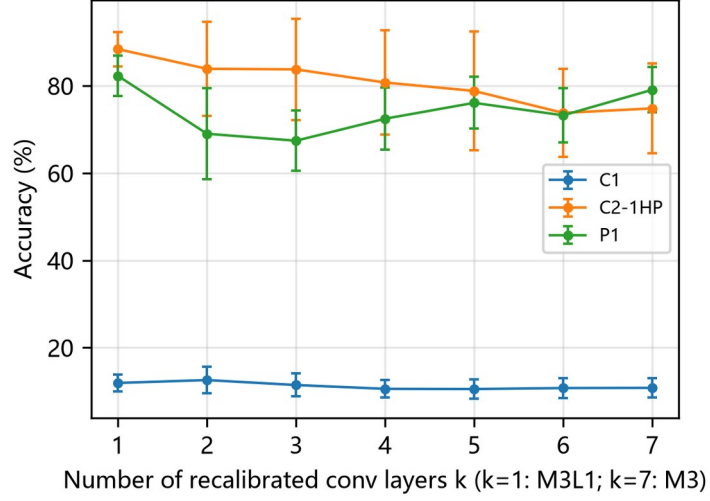


Figure S3. Accuracy versus the number of recalibrated layers k on the development targets (mean $\pm$ std over 5 seeds); the data are those of Table 7 of the paper.

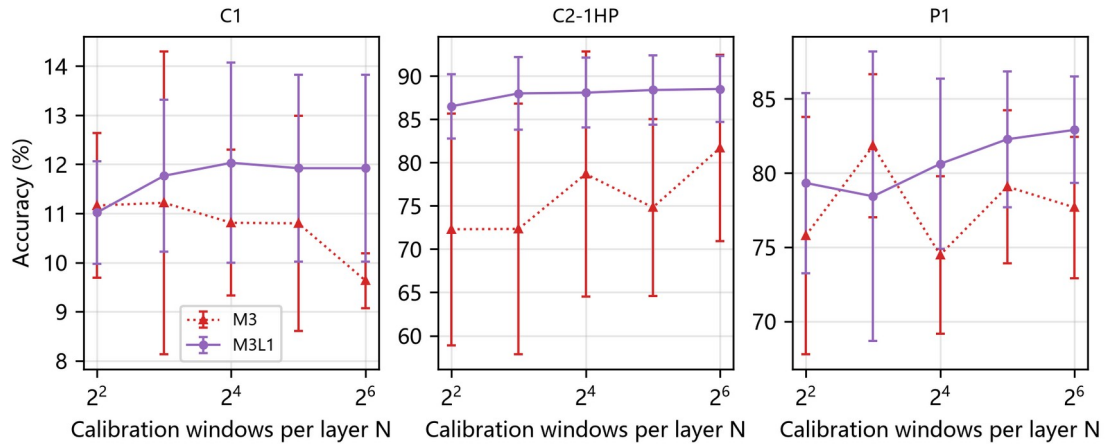


Figure S4. Accuracy versus the number of calibration windows per layer N on the development targets (mean $\pm$ std over 5 seeds).

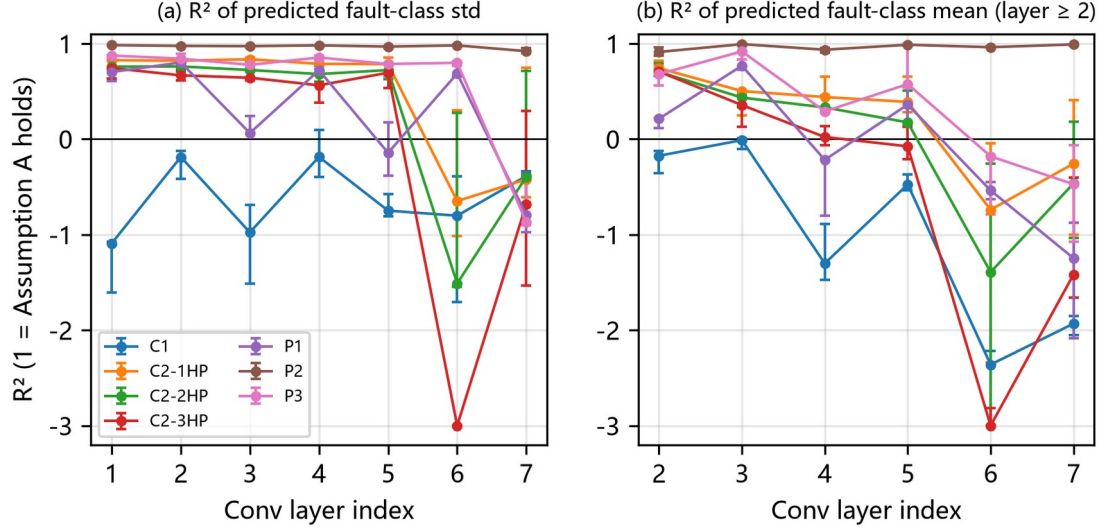


Figure S5. Test of Assumption A: R^2 of the predicted fault-class standard deviation (a) and mean (b, from layer 2) per layer (median of 5 seeds; bars show the interquartile range; values below -3 are drawn at the lower bound).

S5. M1L1 control on all targets

M1L1 recalibrates only the first layer, like M3L1, but aligns to the all-class anchor. It uses the same source models, the same 32 calibration windows per target and seed and the same integer update as M3L1; the recomputed M3L1 values are identical to those of Table 5 of the paper.

Table S4. M1L1 versus M3L1 on all targets (accuracy, %, mean $\pm$ std over 5 seeds). D: development; H: held-out.

Target	M0	M1L1	M3L1
C1 (D)	18.9 $\pm$ 4.3	21.3 $\pm$ 3.2	11.9 $\pm$ 1.9
C2-1HP (D)	90.4 $\pm$ 3.0	46.4 $\pm$ 15.6	88.4 $\pm$ 4.0
P1 (D)	46.8 $\pm$ 8.0	72.4 $\pm$ 4.3	82.3 $\pm$ 4.6
C2-2HP (H)	88.3 $\pm$ 4.1	44.0 $\pm$ 13.0	84.5 $\pm$ 5.7
C2-3HP (H)	83.5 $\pm$ 4.3	47.4 $\pm$ 13.4	83.3 $\pm$ 5.8
P2 (H)	99.8 $\pm$ 0.1	73.3 $\pm$ 2.6	99.8 $\pm$ 0.1
P3 (H)	75.1 $\pm$ 5.5	66.2 $\pm$ 4.7	80.0 $\pm$ 6.2
Mean, all 7	71.8	53.0	75.7
Mean, held-out 4	86.7	57.7	86.9

Over all 35 (target, seed) pairs M3L1 is 22.8 points better than M1L1 (30 wins, 5 losses, all on C1; two-sided Wilcoxon $p = 4.9 \times 10^{-6}$ with the normal approximation used for Table 6 of the paper, not corrected for multiple comparisons); over the 20 held-out pairs it is 29.2 points better (20 wins). With the four held-out targets as units M3L1 wins on all four (exact Wilcoxon $p = 0.125$, the smallest value possible with four units). In the first layer the means of the two anchors almost coincide (Table 8 of the paper), but the ratio of their standard deviations per channel ranges from 0.30 to 2.34 over all sources and seeds (median 0.87–1.19 per source), so M1L1 rescales the healthy data with a wrong gain. Without a shift (protocol of Table 4 of the paper), M1L1 gives $56.80 \pm 19.41\%$ (C1 source), $59.95 \pm 21.62\%$ (C2 source) and $78.55 \pm 3.52\%$ (PU source), whereas M3L1 gives $99.85 \pm 0.09\%$, $99.54 \pm 0.25\%$ and $99.87 \pm 0.12\%$.